CodeAlpha_Data_Visualization

Task 3

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

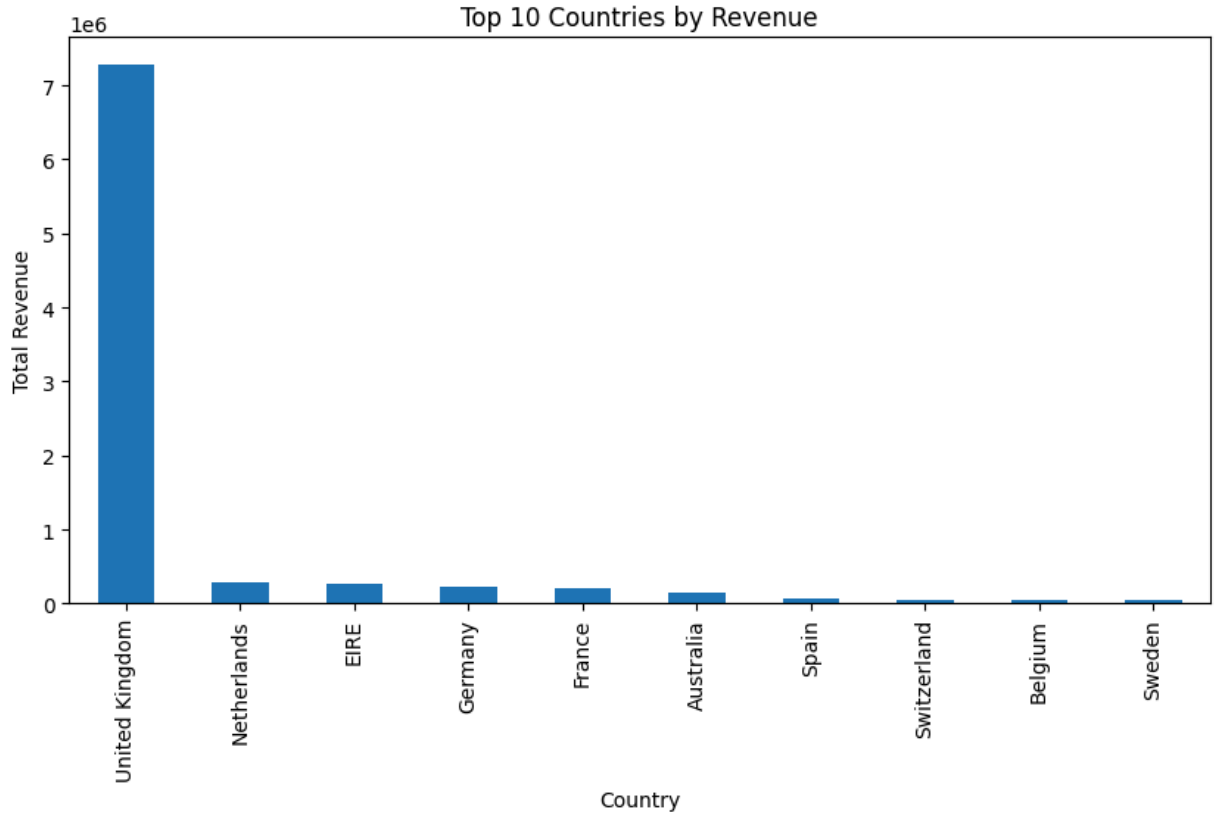
```
df = pd.read_excel("Online Retail Data Set after data cleaning.xlsx")

print(df.head())
print(df.info())
print(df.describe())
```

```
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

          InvoiceDate  UnitPrice  CustomerID         Country
0 2010-12-01 08:26:00       2.55     17850.0  United Kingdom
1 2010-12-01 08:26:00       3.39     17850.0  United Kingdom
2 2010-12-01 08:26:00       2.75     17850.0  United Kingdom
3 2010-12-01 08:26:00       3.39     17850.0  United Kingdom
4 2010-12-01 08:26:00       3.39     17850.0  United Kingdom
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 536641 entries, 0 to 536640
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    536641 non-null  object
 1   StockCode    536641 non-null  object
 2   Description  535187 non-null  object
 3   Quantity     536641 non-null  int64
 4   InvoiceDate  536641 non-null  datetime64[ns]
 5   UnitPrice    536641 non-null  float64
 6   CustomerID   401604 non-null  float64
 7   Country      536641 non-null  object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 32.8+ MB
None
            Quantity                    InvoiceDate      UnitPrice  \
count  536641.000000                         536641  536641.000000
mean        9.620029  2011-07-04 08:57:06.087421952       4.632656
min    -80995.000000            2010-12-01 08:26:00  -11062.060000
```

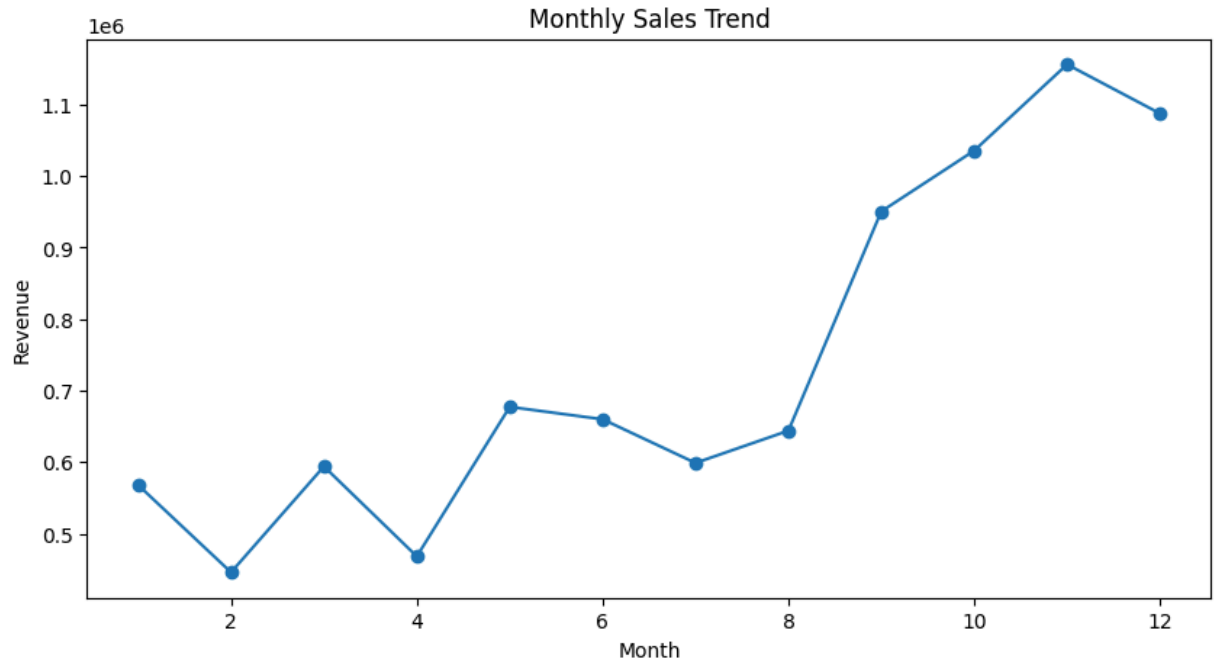
```
25%         1.000000            2011-03-28 10:52:00       1.250000
50%         3.000000            2011-07-19 14:04:00       2.080000
75%        10.000000            2011-10-18 17:05:00       4.130000
max     80995.000000            2011-12-09 12:50:00   38970.000000
std       219.130156                            NaN      97.233118

          CustomerID
count  401604.000000
mean    15281.160818
min     12346.000000
25%     13939.000000
50%     15145.000000
75%     16784.000000
max     18287.000000
std      1714.006089
```

```
# Remove missing values
df = df.dropna()

# Remove negative quantities
df = df[df['Quantity'] > 0]

# Create TotalPrice column
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']

# Convert date
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

# Extract month
df['Month'] = df['InvoiceDate'].dt.month
```

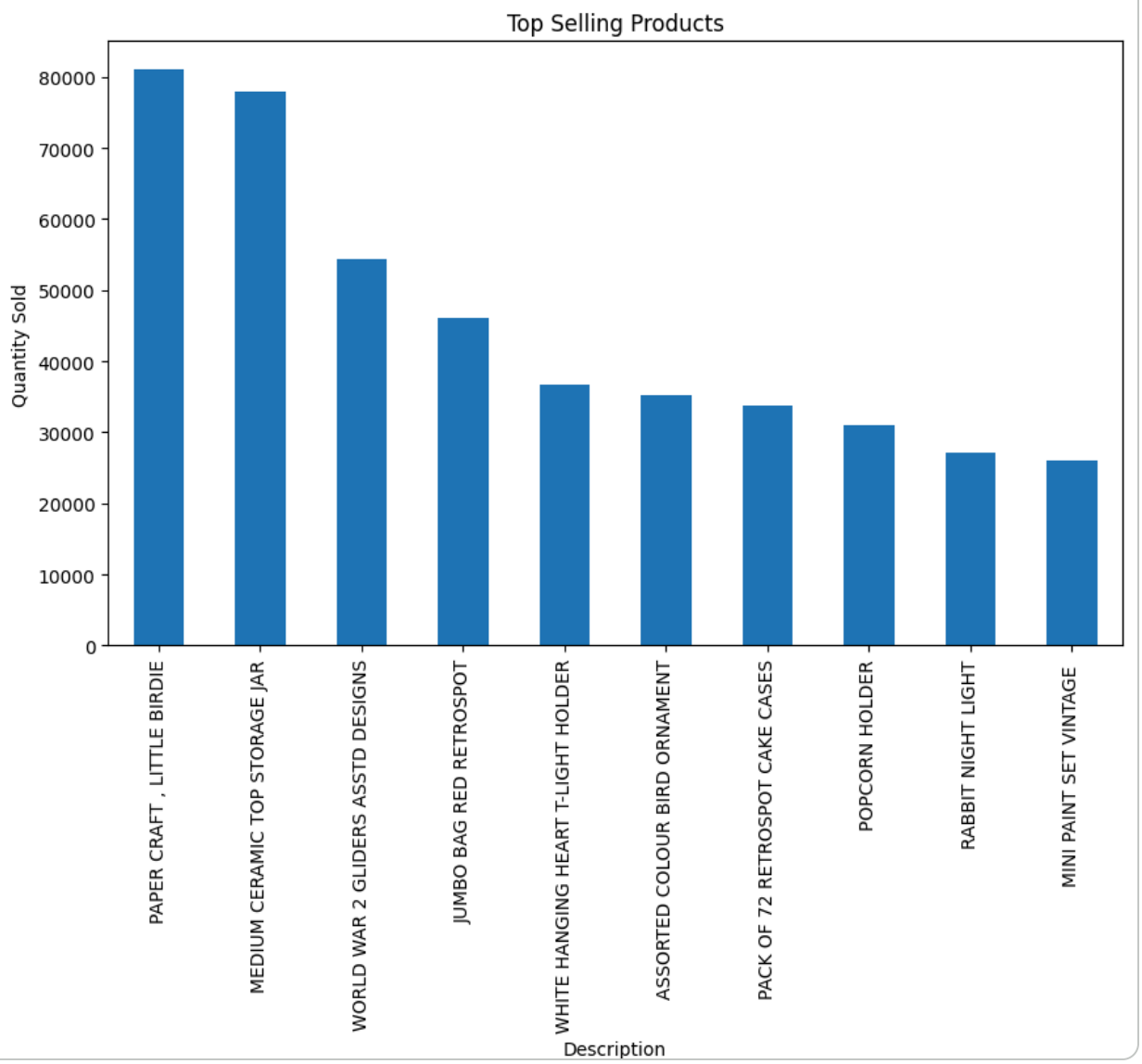
```
country_sales = df.groupby("Country")["TotalPrice"].sum().sort_values(ascending

plt.figure(figsize=(10,5))
country_sales.plot(kind="bar")
plt.title("Top 10 Countries by Revenue")
plt.ylabel("Total Revenue")
plt.show()
```



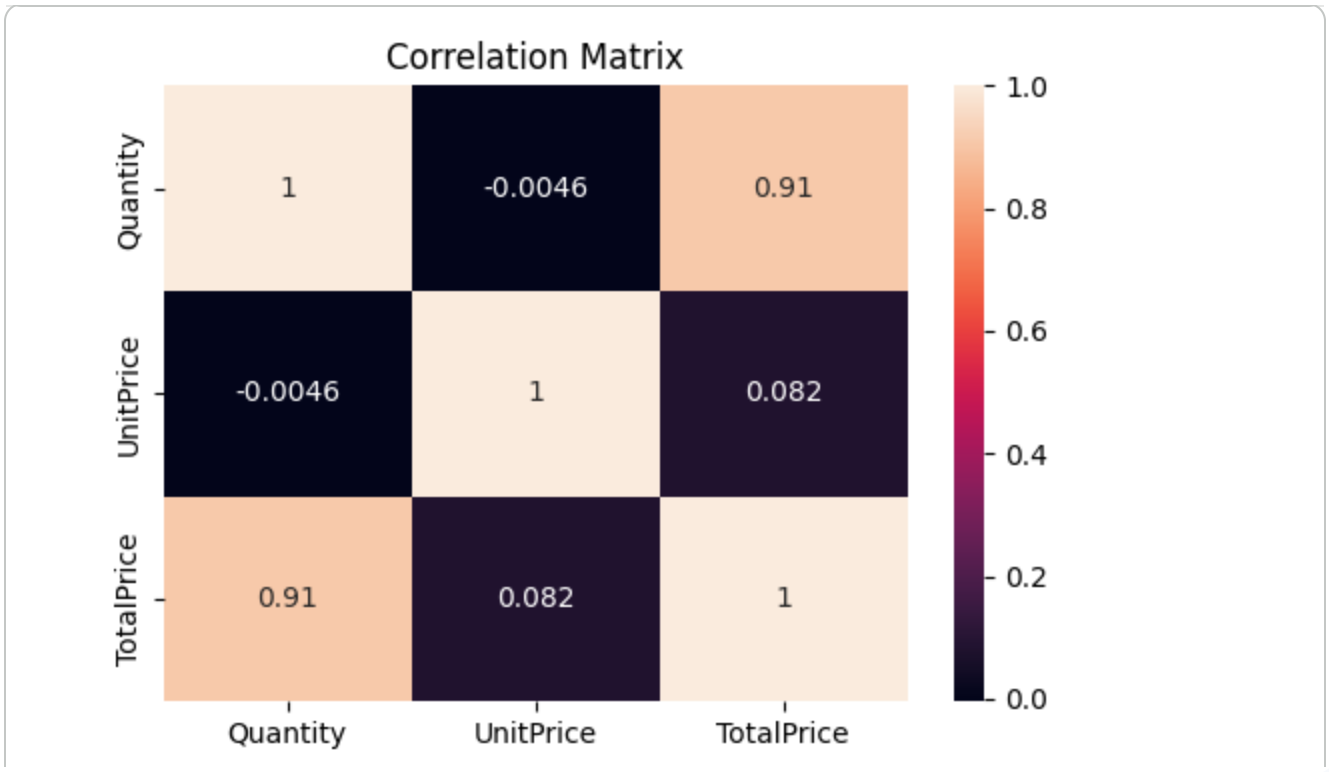

```
monthly_sales = df.groupby("Month")["TotalPrice"].sum()

plt.figure(figsize=(10,5))
monthly_sales.plot(marker="o")
plt.title("Monthly Sales Trend")
plt.ylabel("Revenue")
plt.show()
```



```
top_products = df.groupby("Description")["Quantity"].sum().sort_values(ascendir

plt.figure(figsize=(10,6))
top_products.plot(kind="bar")
plt.title("Top Selling Products")
plt.ylabel("Quantity Sold")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.heatmap(df[['Quantity','UnitPrice','TotalPrice']].corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```


Correlation Matrix
Quantity
UnitPrice
TotalPrice
1
-0.0046
0.91
-0.0046
1
0.082
0.91
0.082
1
Quantity
UnitPrice
TotalPrice
1.0
0.8
0.6
0.4
0.2
0.0